

3.2 WAVES P2&P3 QUESTIONS

Compiled by: R. Muga

Q1: NOV 2002 P2

- 4/5 (a) (i) What do you understand by *diffraction* of waves?

[2]

- (ii) Demonstrate your appreciation of diffraction by completing Fig. 4.1, showing the wave fronts after passing the following obstacle.

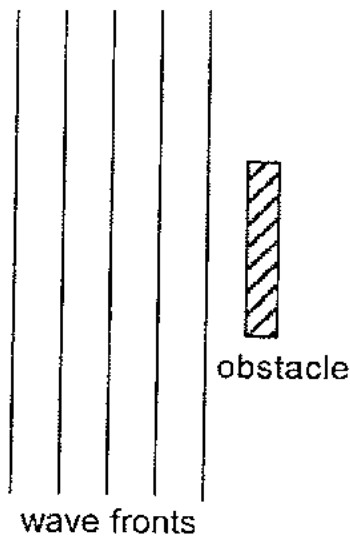


Fig. 4.1

[2]

- (b) Fig. 4.2 shows wave fronts after passing through a narrow and a wide gap positioned on the left-hand end of the pattern.

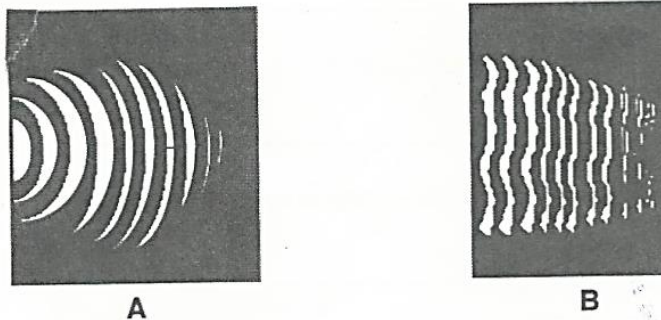


Fig. 4.2

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- (i) State which one of **A** or **B** is the pattern of wave fronts from a narrow gap.

[1]

- (ii) Give a reason for your choice.

[2]

- (c) Why is diffraction of sound waves a common phenomenon but diffraction of light waves is not?

[3]

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Q2:JUN 2008 P2

- 4 (a) Distinguish between a transverse wave and a longitudinal wave. Give one example for each.

[4]

- (b) In an experiment to demonstrate interference of light waves a student made the following observations:

1. *light from two separate light bulbs cannot produce an interference pattern*
2. *if light from an illuminated single slit falls on a double slit gives an interference pattern*

Explain the observations.

1.

2.

[2]

- (c) State the main advantage of using a fibre endoscope in medicine.

[1]

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Q3: NOV 2009 P2

4 (a) Explain the following terms:

(i) *critical angle*

(ii) *total internal reflection*

[2]

(b) Give **one** application of *total internal reflection*.

[1]

(c) Suggest why the sky is blue during daylight.

[2]

Q4: NOV 2012 P2

4 (a) Give **two** similarities and **two** differences between longitudinal and transverse waves.

similarities:

1.

2.

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differences:

1. _____

2. _____

[4]

- (b) Calculate the period, T , of an electromagnetic wave, with a wavelength of 700 nm, travelling in a vacuum.

$T =$ _____ [2]

Q5:NOV 2015 P2

- 4 (a) Explain the term *total internal reflection*.

[2]

- (b) Give four advantages of fibre optic cables over metal cables.

1. _____

2. _____

3. _____

4. _____

[4]

Q6:NOV 2018 P2

- 3 Fig. 3.1 shows plane sound waves passing through a small aperture.

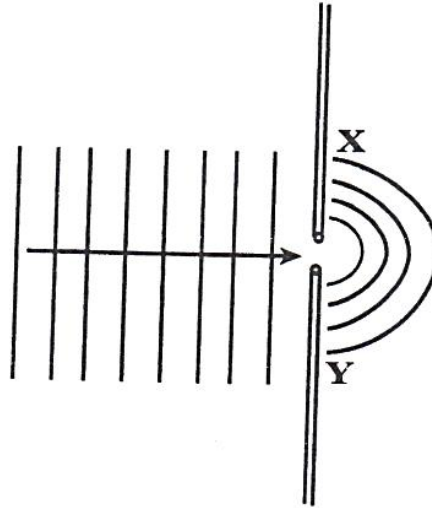


Fig.3.1

Waves are detected at points X and Y.

- (a) Name the phenomenon illustrated in Fig. 3.1.

[1]

- (b) The small aperture is widened almost five fold.

Sketch the plane sound waves after they have just passed the widened aperture.

- 4 A 2 x 2 voxel with intensities shown in Fig.4.1 (a) is used in CT scanning to obtain the fixed pattern in Fig.4.1(b)

1	5
3	8

Fig. 4.1 (a)

20	32
26	41

Fig. 4.1 (b)

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- (a) Determine the background intensity.

[1]

- (b) Obtain the pattern in **Fig.4.1 (a)** using the final pattern in **Fig.4.1(b)**

[2]

- (c) State **one** advantage and **one** disadvantage of using CT scanning.

Advantage _____

Disadvantage _____

[2]

Q7:JUN 2019 P2

- 2 (a) Fig. 2.1 shows a device used for detecting the level of acid in a battery.

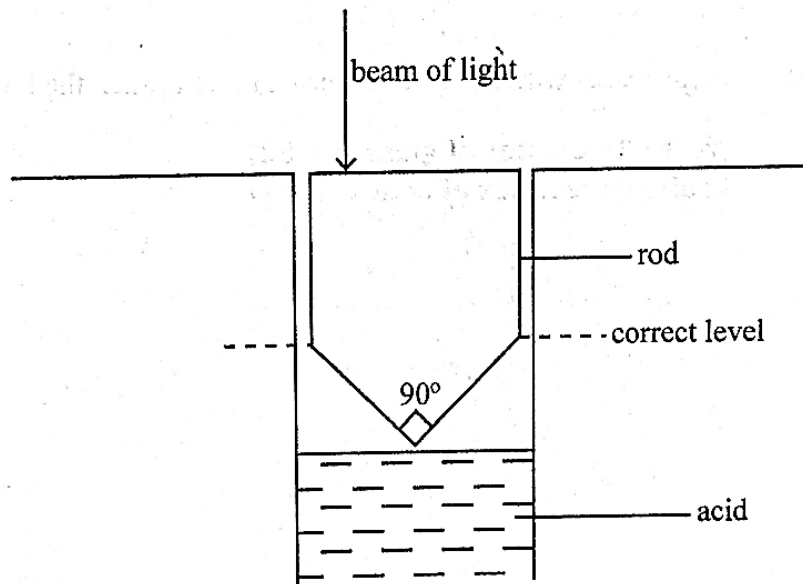


Fig. 2.1

A glass rod with its end shaped as a right angled prism dips into the acid when the level is correct. When the acid level is too low, the end of the rod is in air.

- (i) A beam of light is shone on one side of the rod and a strong reflected beam is observed emerging upwards from the other side when the acid level is too low.

Explain this effect.

- (ii) Predict what will happen to the beam of light when the level is correct.

[refractive index of glass = 1.52]
[refractive index of acid = 1.35]

Q8:2018 SPEC P2

3. Fig. 3.1 shows a ray incident on a glass-air boundary. The glass-air and air-glass boundaries are parallel to each other.

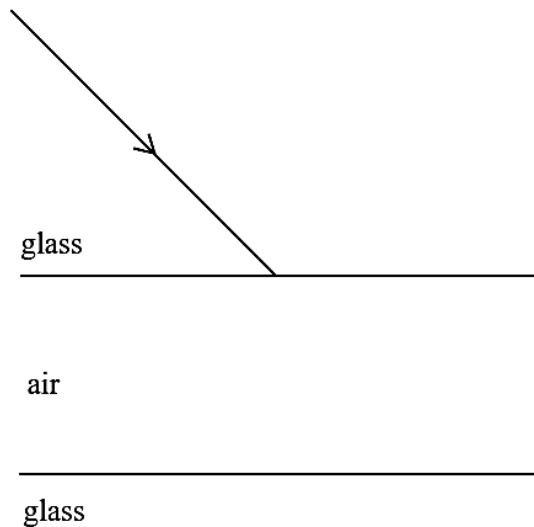


Fig. 3.1

- (a) (i) Complete the path of the ray as it passes through the air into glass.
(ii) Determine the angle to the horizontal at which the ray emerges from air entering glass if the angle of incidence in glass is 40° .

[5]

- (b) Derive the equation $n = \frac{1}{\sin c}$.

4. Fig. 4.1(a) and Fig 4.1(b) shows an X-ray beam from an anode target.

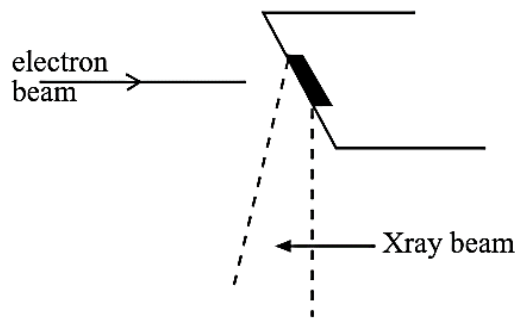


Fig.4.1(a)

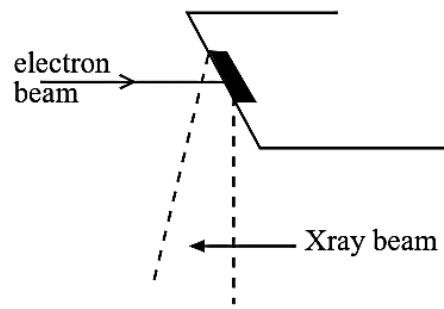


Fig 4.1(b)

- (a) (i) State the figure which shows the X-ray beam preferred in X-ray imaging.

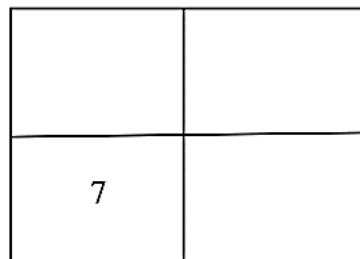
- (ii) Explain your choice in part (a)(i).

_____ [2]

- (b) Explain why a patient is given a barium meal before an X-ray image of the stomach is taken.

_____ [3]

The number 7 is written in a voxel as shown.



- (c) State what the number 7 represents.

_____ [1]

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- (d) State the advantage of the image obtained using CT scanning over the image obtained using X-rays.

[1]

Q9:JUN 2007 P3

- 3 (a) (i) State the laws of refraction.
- (ii) 1. Explain what is meant by total internal reflection.
2. State the conditions in which it occurs.
3. Give one application of total internal reflection. [6]
-  (b) Calculate the angular separation of the red and violet rays which emerge from a 60° glass prism when a ray of white light is incident on the prism at an angle of 45° . The glass has refractive indices of 1.63 and 1.66 for red and violet light respectively. [5]
- (c) (i) Define plane polarisation of electromagnetic radiation.
- (ii) Describe how plane polarised light can be produced and detected. [4]

Q10:NOV 2007 P3

- 3 (a) State the *Laws of refraction*. [2]
- (b) When light passes from glass into air, it bends away from the normal.
- (i) Explain using ray diagrams how this fact leads to total internal reflection.
- (ii) Describe the use of total internal reflection in fibre optics.
- (iii) Give one advantage of fibre optics in communication. [7]
- (c) (i) Distinguish between *transverse* and *longitudinal* waves and give an example of each.
- (ii) Explain what is meant by the term diffraction as applied to wave motion.
- (iii) Calculate the wavelength associated with a sprinter of mass 60.0 kg who runs through an open door at a velocity of 10.0 ms^{-1} .
- (iv) Explain why no diffraction pattern is observed in (c)(iii). [10]
- (d) State one phenomenon which is associated with transverse waves only. [1]

Q11:JUN 2008 P3

- 3 (a) Using diagrams explain the meaning of the terms *total internal reflection* and *critical angle C*. [5]
- (b) Fig. 3.1 shows light entering an optical fibre made up of a glass core surrounded by a glass cladding. The refractive index, n_2 , of the cladding is less than that of the core, n_1 .

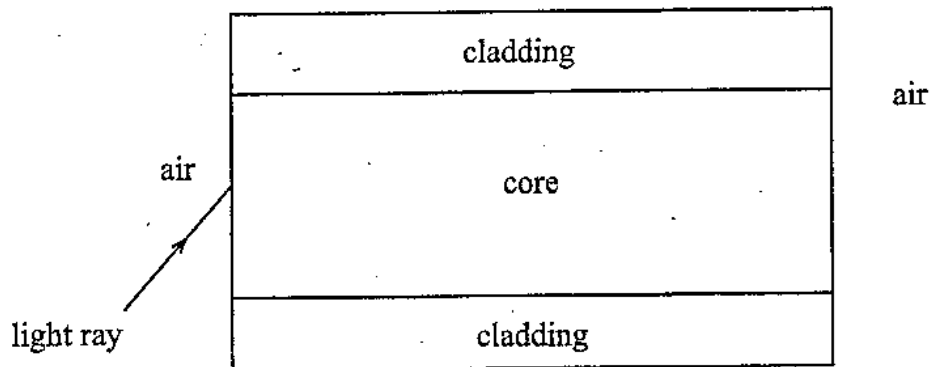


Fig. 3.1

- (i) Explain how the optical fibre transmits a light signal from one end to another.
- (ii) Show that $\sin C = \frac{n_2}{n_1}$ [5]
- (c) The refractive indices of the core and cladding of an optical fibre are 1.6 and 1.4 respectively.
- (i) Calculate the critical angle at the core-cladding interface.
- (ii) Deduce that the minimum angle of incidence, i , for which all the light is totally reflected at the core-cladding interface is given by
- $$\sin i = \sqrt{n_1^2 - n_2^2}, \text{ where } n_{\text{air}} = 1.$$
- Hence calculate the minimum angle of incidence in the air that will cause the incident ray to arrive at the core-cladding interface at the critical angle. [6]
- (d) (i) Suggest **two** uses of the cladding material.
- (ii) State **two** advantages of fibre optics transmission. [4]

Q12:JUN 2011 P3

- 2 (b) A ray of light enters a 60° triangular prism and is just totally internally reflected when it meets the opposite face of the prism as shown in Fig. 2.1.

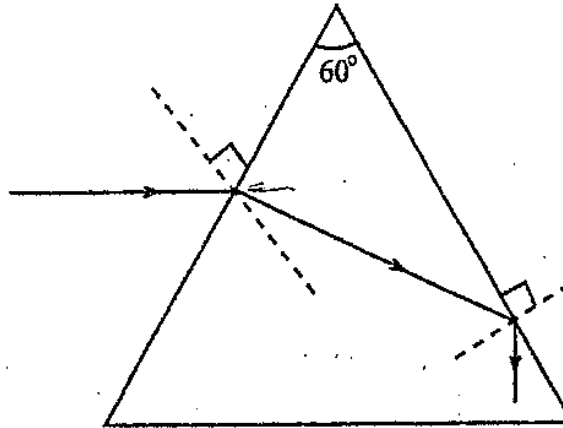


Fig. 2.1

- (i) State the law of refraction relating the angle of incidence to the angle of refraction.
- (ii) Given that the refractive index of the prism is 1.5, calculate the angle of incidence of the ray of light on the first face.

[6]

Q13:JUN 2013 P3

- 4 (a) (i) Imaging is one of the uses of X-rays in Medicine.
- Give **two** kinds of cases where X-rays would produce the best images.
- (ii) In medical diagnosis, X-rays are passed through a filter before they pass through a patient. After the patient, the rays pass through a grid before they reach the X-ray film as shown in Fig.4.1

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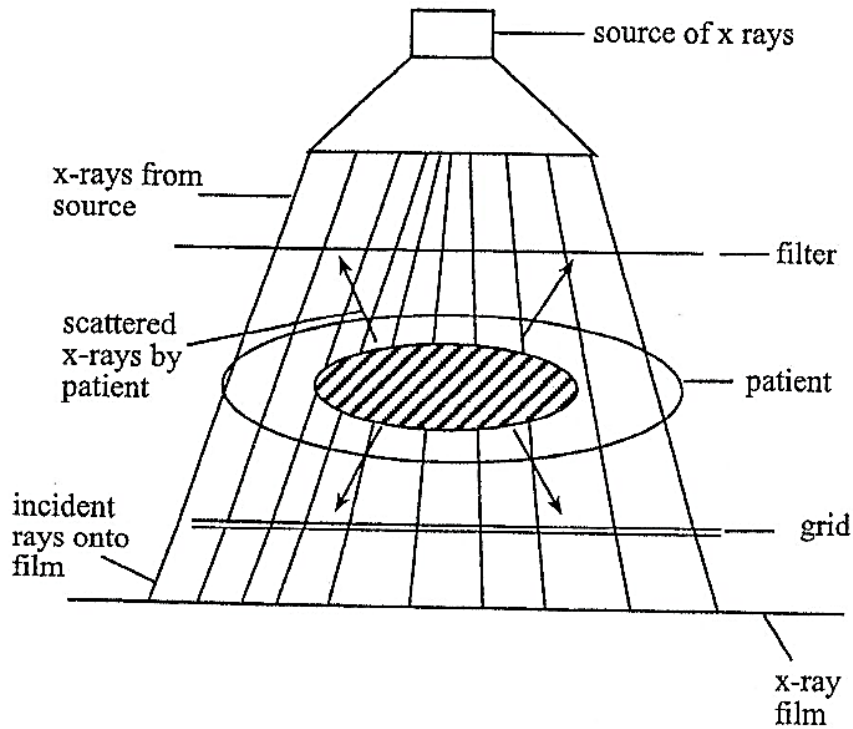


Fig. 4.1

State and explain the advantages of this procedure regarding the patient and quality of the image,

[8]

(b) Lasers are used in medical treatment in a variety of ways.

(i) State any **one** common medical use of lasers.

(ii) Briefly explain how lasers are used in (i).

[2]

Q14:JUN 2014 P3

4 (a) State the **two** laws of reflection of light.

[2]

(b) Fig. 4.1 shows a ray of monochromatic light reaching a glass prism of refraction index 1.55.

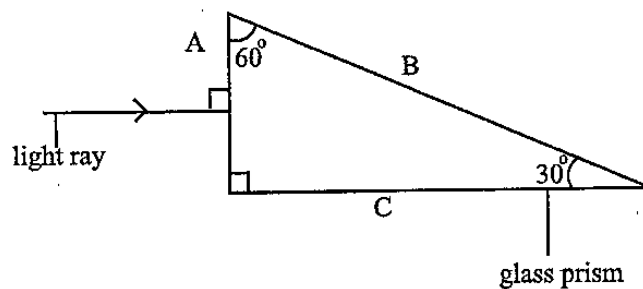


Fig. 4.1

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- (i) Calculate the critical angle for this glass and air.
- (ii) Copy Fig. 4.1 and complete the path of the ray as it enters face A, moves to face B, then to face C, and after face C.

Give a brief explanation for the path you have drawn.

- (iii) Draw a diagram to show how total internal reflection occurs in fibre optics transmission. [6]

- (c) In Zimbabwe copper cables are being replaced by fibre optic cables along the major high ways.

Explain why this is good for the country. [2]

Q15:JUN 2015 P3

- 1 (c) (i) State the **three** properties of an image formed by a plane mirror.

- (ii) A ray of light enters glass from air. The angle of incidence is 40° and that of refraction is 26° .

Calculate the critical angle of glass.

[6]

- 4 (a) Define *total internal reflection*.

[2]

- (b) (i) With the aid of a diagram, describe how light travels through an optical fibre shown in Fig.4.1.

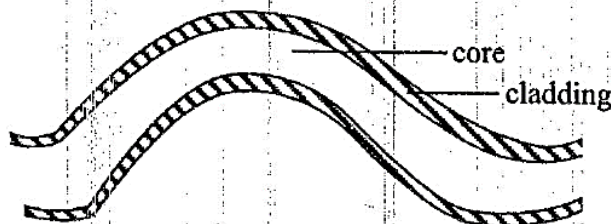


Fig.4.1

- (ii) Explain why optical fibres are more preferred than electrical cables.

[6]

- (c) Explain why the red colour is used in breakdown triangle reflectors.

[2]

Q16:JUN 2017 P3

- 1 (d)**
- (i) Define *critical angle*.
 - (ii) Explain why the critical angle is never reached when light moves from an optically less dense medium to an optically denser medium.
 - (iii) Suggest why optical cables are made up of very thin fibres but not one solid thick rod. [4]

Q17:NOV 2018 P3

- (c)
- (i) State any one property of X-ray radiation.
 - (ii) Estimate the wavelength of X-ray radiation.
 - (iii) With the aid of a well labelled diagram, describe how X-rays are produced in the tube.
 - (iv) Describe the energy changes in the x-ray production. [8]

Q18:2018 SPEC P3

- 2 (c)** An X-ray beam of intensity $8.00 \times 10^5 \text{ W m}^{-2}$ passes through an aluminium filter and linear attenuation coefficient μ of 250 m^{-1} . The filter allows 10% of the X-ray radiation to pass through it to a patient whose bone under examination has a thickness of 4.0 cm and attenuation coefficient of 1.05 m^{-1} .

- (i) Calculate the
 - 1. thickness of aluminium,
 - 2. intensity of the X-ray radiation transmitted by the bone.
- (ii) Suggest the advantages of passing the X-rays through the filter before passing it through the patient.
- (iii) Explain how an anti-scatter grid improves the X-ray image produced. [8]

3.2 WAVES P2&P3 QUESTIONS

Compiled by: R. Nuzo

- (d) (i) Outline how X-rays are used to build up the images produced in a CT scan.
- (ii) Explain why CT scanning requires a powerful computer. [4]

Q19:JUN 2019 P3

- (c) (i) Describe the main principles of CT scanning.
- (ii) State the advantages of CT scanning over X-ray imaging. [8]

Q20:2006 SPEC P2

- 3 X-rays are produced when high speed electrons are stopped by a metal target. Fig. 3.1 shows two x-rays spectra.

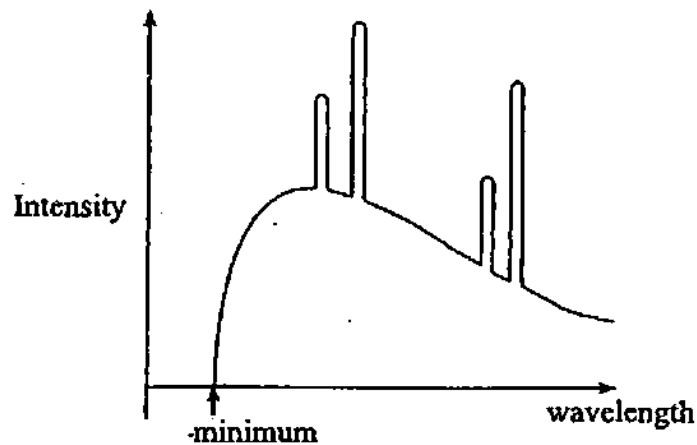


Fig. 3.1

- (a) Suggest an explanation for the sharp peaks on the spectrum.

[4]

3.2 WAVES P2&P3 QUESTIONS

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- (b) On Fig. 3.1 add a curve to show the spectrum produced by use of a smaller accelerating potential difference on the same metal target.

[2]

- (c) Explain how the minimum wave length $\lambda_{\min}$ is obtained.

[2]

- (d) The electrons are accelerated by a p.d of 40 kV.

- (i) Calculate the kinetic energy of one accelerated electron.

kinetic energy = _____

- (ii) Hence find the minimum wavelength, $\lambda_{\min}$.

$\lambda_{\min} =$ _____ [4]

- (e) Fig. 3.2 shows a ray of sunlight incident onto a spherical drop of water in space.

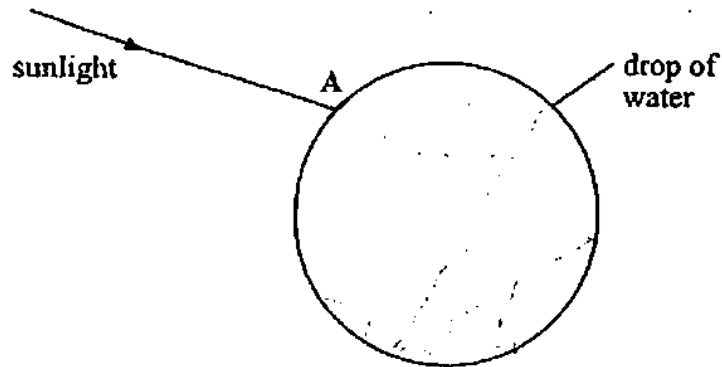


Fig. 3.2

Complete the diagram in Fig. 3.2, showing the paths of red, green and violet rays from point A through the drop of water to produce a primary rainbow. Label the paths accordingly.

[4]

Q21:JUN 2019 P2

- 5 (b) (i) State what is meant by *hardness of an X ray beam*.

- (ii) Fig. 2.2 shows part of an X ray tube.

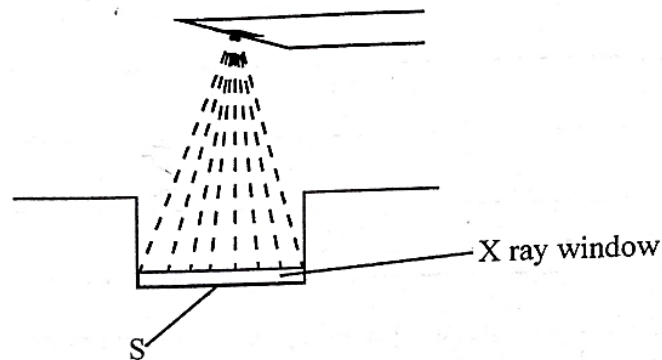


Fig. 2.2

1. Suggest a suitable material for S.

2. State the purpose of S.

3.2 WAVES P2&P3 QUESTIONS

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3. Give **one** method of improving the sharpness of an X ray image.

[4]